PALMO

IoT project report

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Real-life project context

The philosophical foundation of PALMO
rests on a simple, universal human reality:

your hand is always with you.



This project models simple financial system

Where all you need to interact with finances
is your biometry data

Just the scan of palm allows to pay in shops
and deposit money

Edge devices



Point of sale(PoS) device
uses PALM to
acesse balance for purchase
Sends HASH to the topic



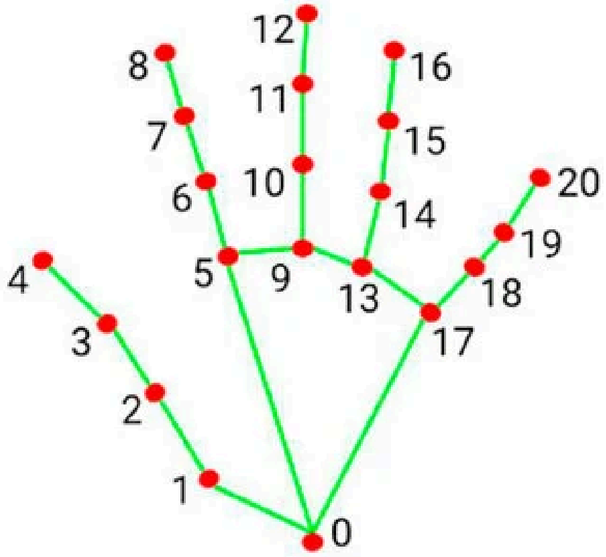
Top-up device
uses PALM to access
account for top-up



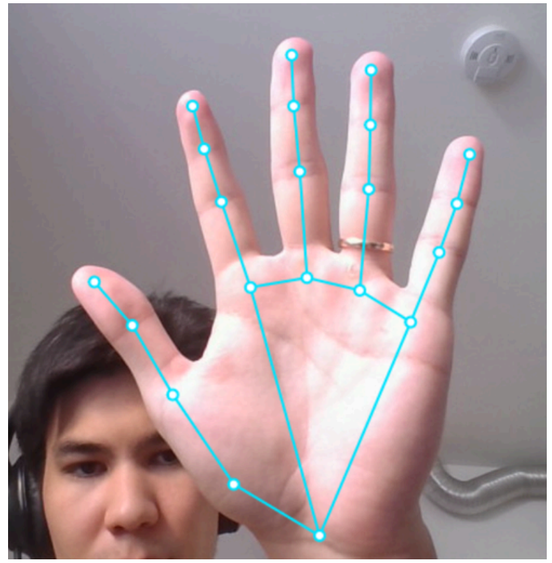
Registrator device
Reads and writes PALM HASH
Creates account with initial balance

MediaPipe Hands

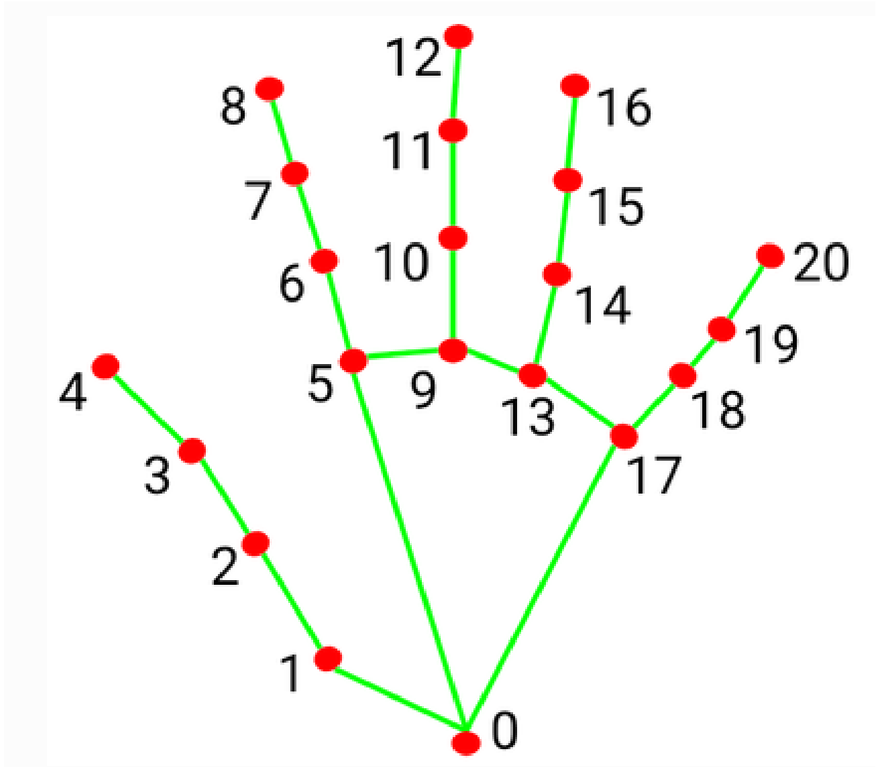
MediaPipe Hands is a high-fidelity hand and finger tracking solution. It employs machine learning (ML) to infer 21 3D landmarks of a hand from just a single frame



0. WRIST
1. THUMB_CMC
2. THUMB_MCP
3. THUMB_IP
4. THUMB_TIP
5. INDEX_FINGER_MCP
6. INDEX_FINGER_PIP
7. INDEX_FINGER_DIP
8. INDEX_FINGER_TIP
9. MIDDLE_FINGER_MCP
10. MIDDLE_FINGER_PIP
11. MIDDLE_FINGER_DIP
12. MIDDLE_FINGER_TIP
13. RING_FINGER_MCP
14. RING_FINGER_PIP
15. RING_FINGER_DIP
16. RING_FINGER_TIP
17. PINKY_MCP
18. PINKY_PIP
19. PINKY_DIP
20. PINKY_TIP



Hand → Hash



By default we get 21 landmarks - points (x, y, z)
z - depth relative to point 0.

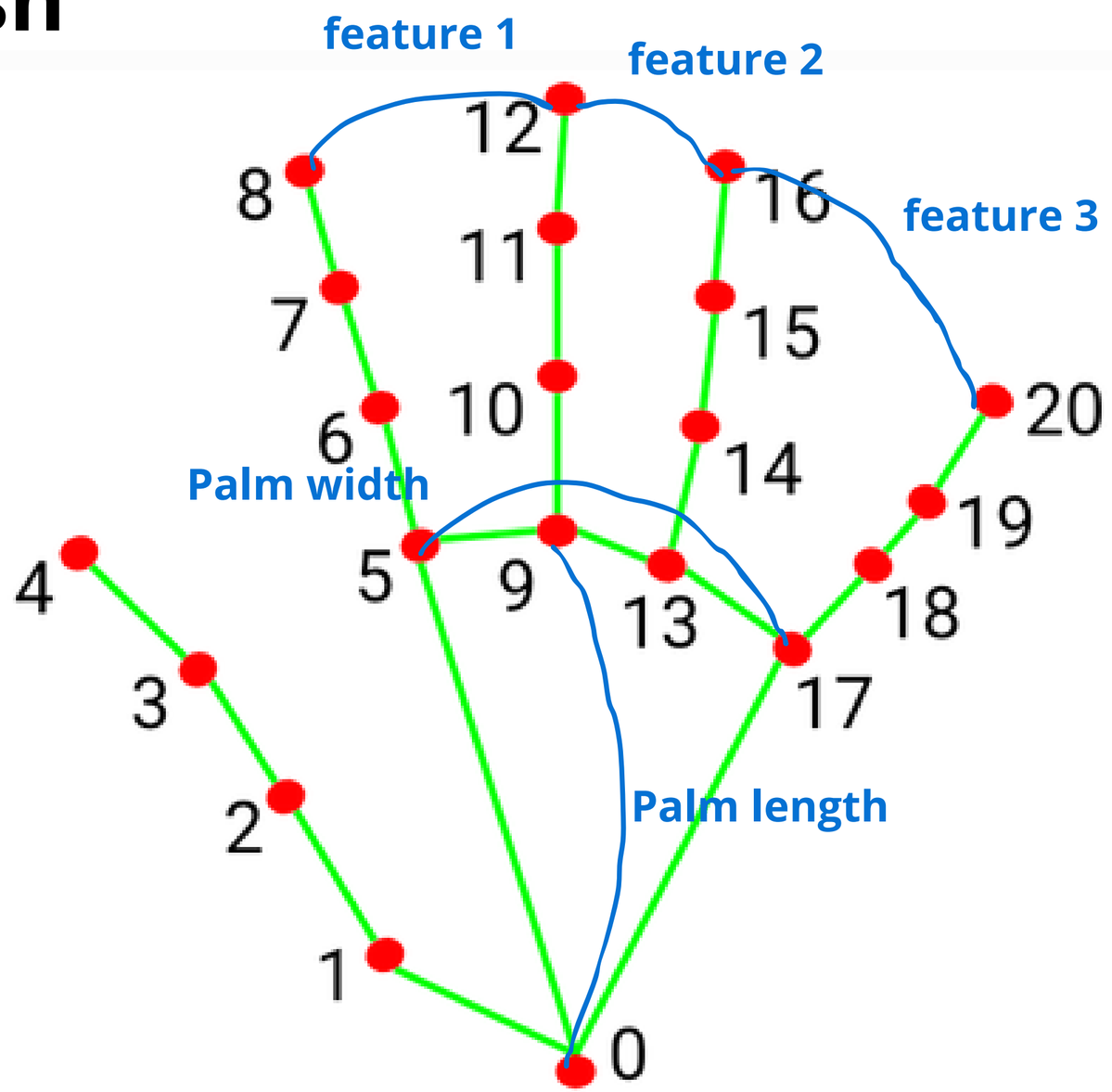
Raw landmarks are very unstable for hash production, as small changes in position will dramaticaly change results of scanning

- Landmarks depend from angle towards camera
- Landmarks depend from distance towards camera
- Landmarks are unique for position, but not for person

Hand →Features → Hash

Features are calculated from landmarks data
and represent more stable and universal data per palm

Feature examples:
1) Palm width - distance between p5 and p17
2) Distance/palm width for different pairs of landmarks
3) Palm length - distance between p0 and p9
4) Ratio Length/Width



Features → Hash

Feature value ([-x, +x])

↓

Average for multiple readings

↓

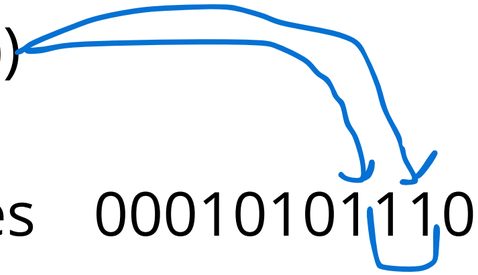
Normalization by pre-specified range (→ [0, 1])

↓

Quantization to level ([0, 1] → (0, 1, 2, 3))

↓

To bits((0, 1, 2, 3) → (00, 01, 10, 11))



↓

Creating bits string from all features 00010101110101000011001

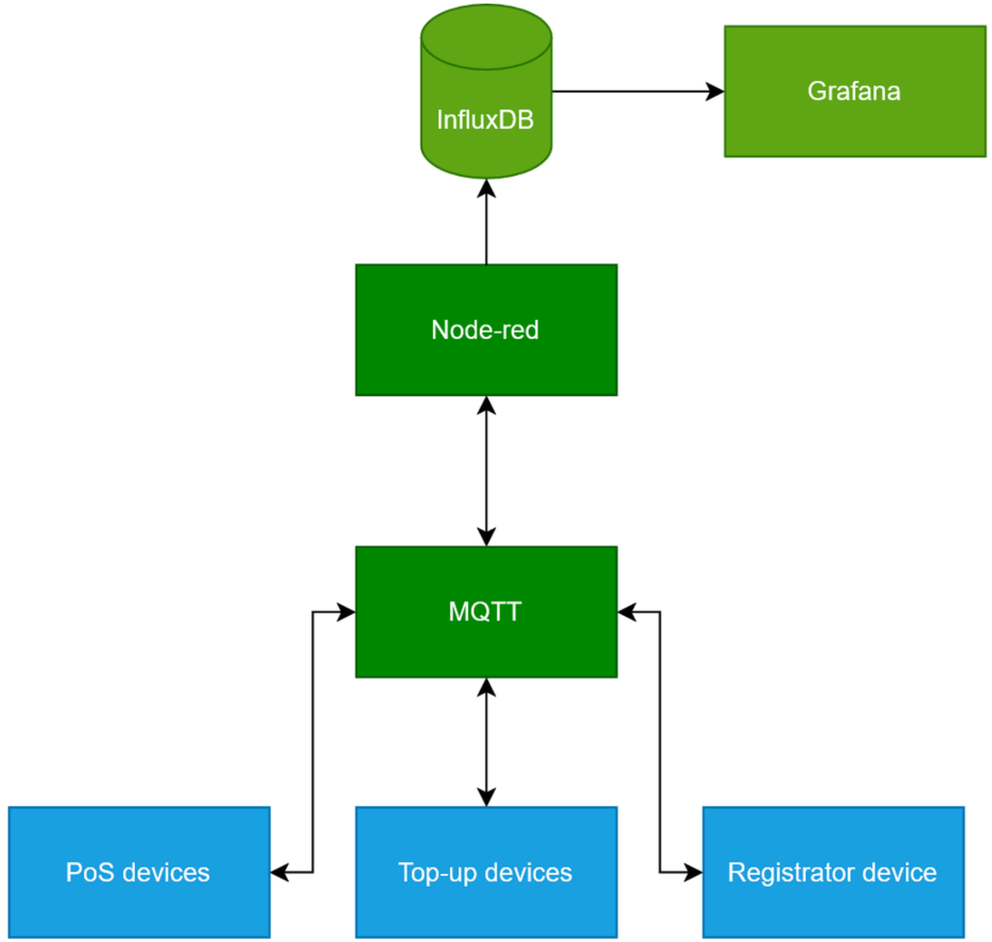
↓

Generating SHA-256 hash string

a4ee16e7741693b4d4e7587875ac219533b271c5a53368122d8e40d3db26ce6

Structure & used techologies

- Edge devices are implemented using **NodeJS**
- Data routing managed using asynchronous publish/subscribe messaging via the **MQTT**
- **Node-RED** is a visual, low-code backend/middleware with logic and connections
- **InfluxDB** acts as an immutable financial ledger (Time-Series DB).
- **Grafana** provides real-time monitoring and analytics visualization



MQTT Topics

palmo/pos/tx - this topic is used by pos device to send data about purchase attempt, expected payload: palm hash, amount, device id, timestamp

palmo/pos/${deviceId}/response - this unique per device topic is used to listen for results of transaction

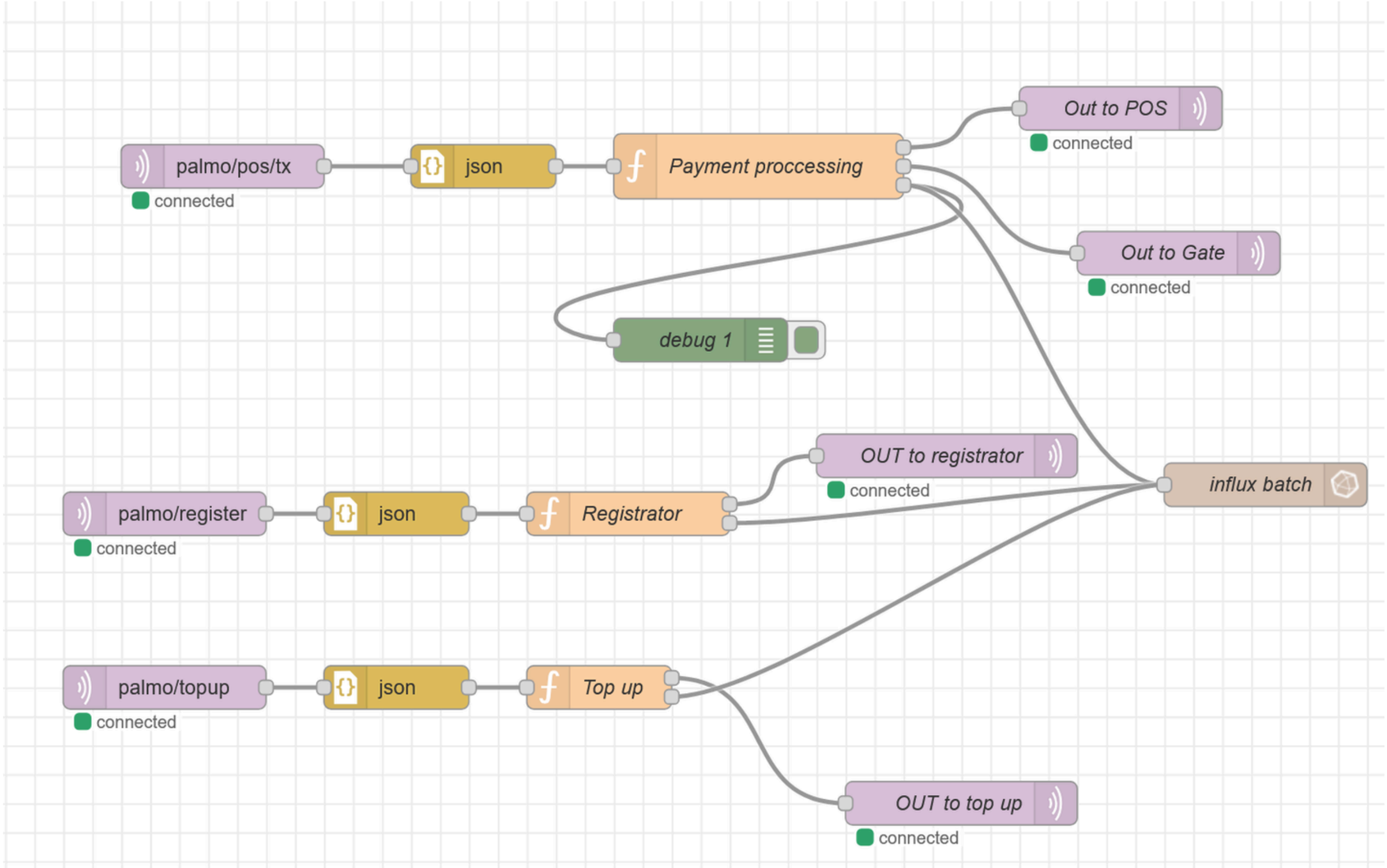
palmo/register - this topic is used to register new user, expected payload: palm hash, initial balance, timestamp

palmo/register/status - this topic is used to listen for registration results

palmo/topup - this topic is by top-up device to send data about new deposit attempt, expected payload: palm hash, amount, device id, timestamp

palmo/topup/${deviceId}/status - this unique per device topic is used to listen for results of top-up

Node-red flow



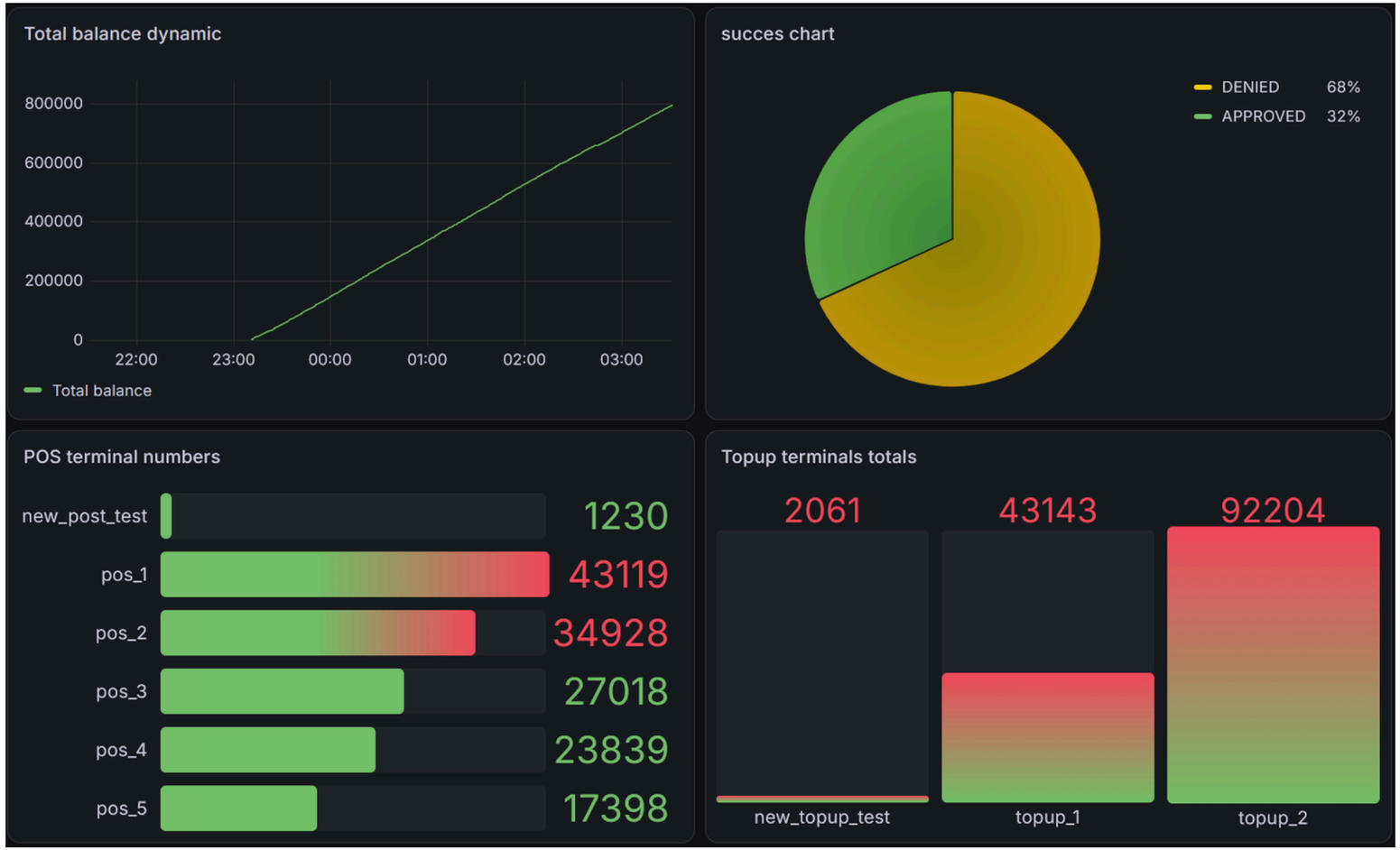
InfluxDB data

- Measurement
  - transaction
  - topup
  - registration
- Palm hash — "fc2fd1422e5118125b1f676bee9019a00eb3c4190ecc110a840998dcf9aeaca9"
- Amount — 145 EUR
- Status
  - Success
  - Denied

Grafana dashboard

We can observe the state of the system
Statistics and perfomance of devices
Dynamics and changes

Here, for example, can be observed total number of money spent through each POS device during given time period



Useful links

GitHub repository
https://github.com/DaurKas/palmo_iot

MediaPipe Hands
https://mediapipe.readthedocs.io/en/latest/solutions/hands.html
https://developers.google.com/edge/mediapipe/solutions/vision/hand_landmarker/web_js